Symplectic topology

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1 Motivation

Symplectic topology is the study of global phenomenon of *symplectic manifolds*, which are a pair (M, ω), where M is a smooth manifold, and ω is a non-degenerate closed 2-form on M. In particular, non-generacy implies the map

$$\begin{aligned} TM &\to T^*M \\ v &\mapsto \omega(v, \cdot) \end{aligned}$$

is an isomorphism.

Example 1.1
$\mathbb{R}^{2n}$ with coordinates $x_1, y_1, \ldots, x_n, y_n$ is a symplectic manifold, with

$$\omega_{std} = \omega_0 = \sum dx_i \wedge dy_i$$

Example 1.2
Let Σ be a surface of genus g. Then any volume form makes Σ_g into a symplectic manifold.

Why study such manifolds? Some motivation from classical mechanics: Say we have a phase space X encoding position and momentum of some particles in a physical system. For example, we have a pendulum with angular coordinate θ, and angular momentum $p = m\dot{\theta}$. The phase space is then $S^1 \times \mathbb{R}$.

Given $H : X \to \mathbb{R}$, encoding the total energy of a configuration, we want a vector field $\mathcal{X}_H$ such that the trajectory $x(t)$ satisfies

$$\dot{x}(t) = \mathcal{X}_H(x(t))$$

It's not enough to know that H is constant on trajectories, i.e. $dH(\mathcal{X}_H) = 0$, since this is a co-dimension 1 condition. Recall if X has symplectic form ω, then we can take $\mathcal{X}_H$ such that

$$\omega(\mathcal{X}_H, \cdot) = dH \tag{1}$$

Indeed, every phase space has such a symplectic form. Note that ω is non-degenerate if and only if eq. (1) has a solution for all H, and ω being closed is that ω is preserved under the flow of $\mathcal{X}_H$.

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Example 1.3 (harmonic oscillator)

In this case, $X = \mathbb{R}^2$, with coordinates p for momentum, q for position. The symplectic form is

$$\mathrm{d}p \wedge \mathrm{d}q = r\mathrm{d}r \wedge \mathrm{d}\theta$$

where (r, θ) is polar coordinates on X. This has Hamiltonian

$$H = \frac{1}{2}(p^2 + q^2) = \frac{1}{2}r^2$$

Hence

$$\mathrm{d}H = r\mathrm{d}r$$

Thus, $\mathcal{X}_H = \frac{\partial}{\partial\theta}$.

1.1 Mathematical perspective

A question to keep in mind is: How does symplectic geometry compare to e.g. Riemannian geometry, complex geometry etc.

In this context, $\mathrm{d}\omega = 0$ is a form of "flatness". Suppose (Y, g) is a Riemannian manifold, i.e. we have $g \in \mathrm{Sym}^2 \mathrm{T}^*Y$ non-degenerate. It is a fact that (Y, g) is locally isometric to $\mathbb{R}^n$ with the Euclidean metric if and only if the curvature F of g vanishes. That is, g is flat.

On the other hand, suppose (Z, J) is an almost complex manifold. That is, $J \in \mathrm{End}(\mathrm{T}Z)$, with $J^2 = -\mathrm{id}_{\mathrm{T}Z}$. One example would be $\mathbb{C}^n$, with $J(z) = iz$. One question is, when is (Z, J) locally isomorphic to $\mathbb{C}^n$? That is, when is (Z, J) a complex manifold. A difficult result in analysis, called the Newlander-Nirenberg theorem, shows that this occurs if and only if the Nijenhuis tensor,

$$\mathcal{N}_J(v, w) = [Jv, Jw] - J([v, Jw] + [Jv, w]) - [v, w]$$

vanishes

Theorem 1.4 (Darboux)**.** A manifold (X, ω) with non-degenerate 2-form ω, is locally modeled on $(\mathbb{R}^{2n}, \omega_{std})$ if and only if $\mathrm{d}\omega = 0$.

Finally, symplectic topology is ubiquitous in research in geometry and topology, e.g. global topology, algebraic geometry and so on.